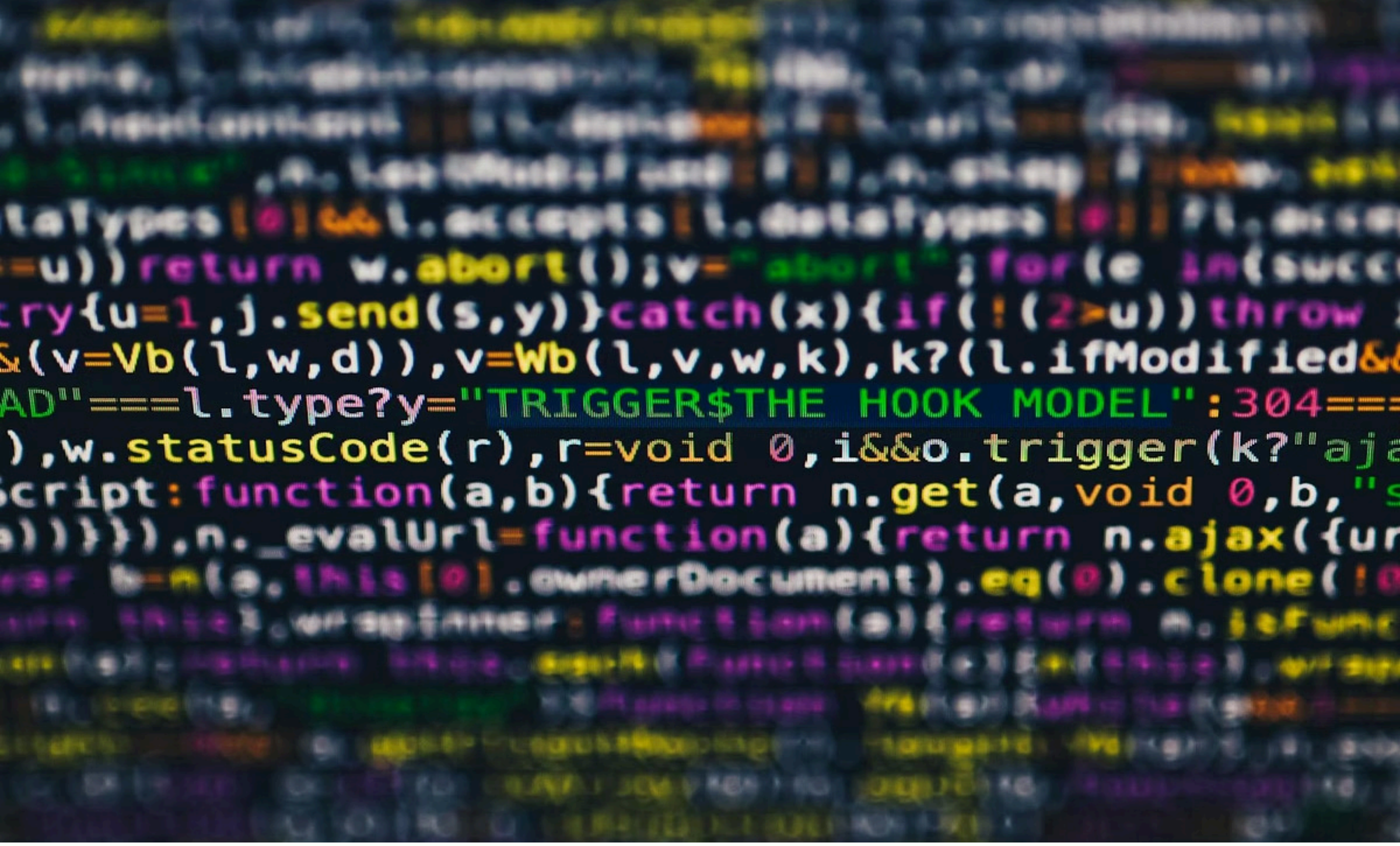


Is the recommendation engine broken?

It thinks it is getting smarter.

But it might slowly be **erasing the catalog.**

Made with GAMMA

The system starts by guessing.

On day one it has no user data. So it ranks products by what it knows: image quality, price, and seller ratings.

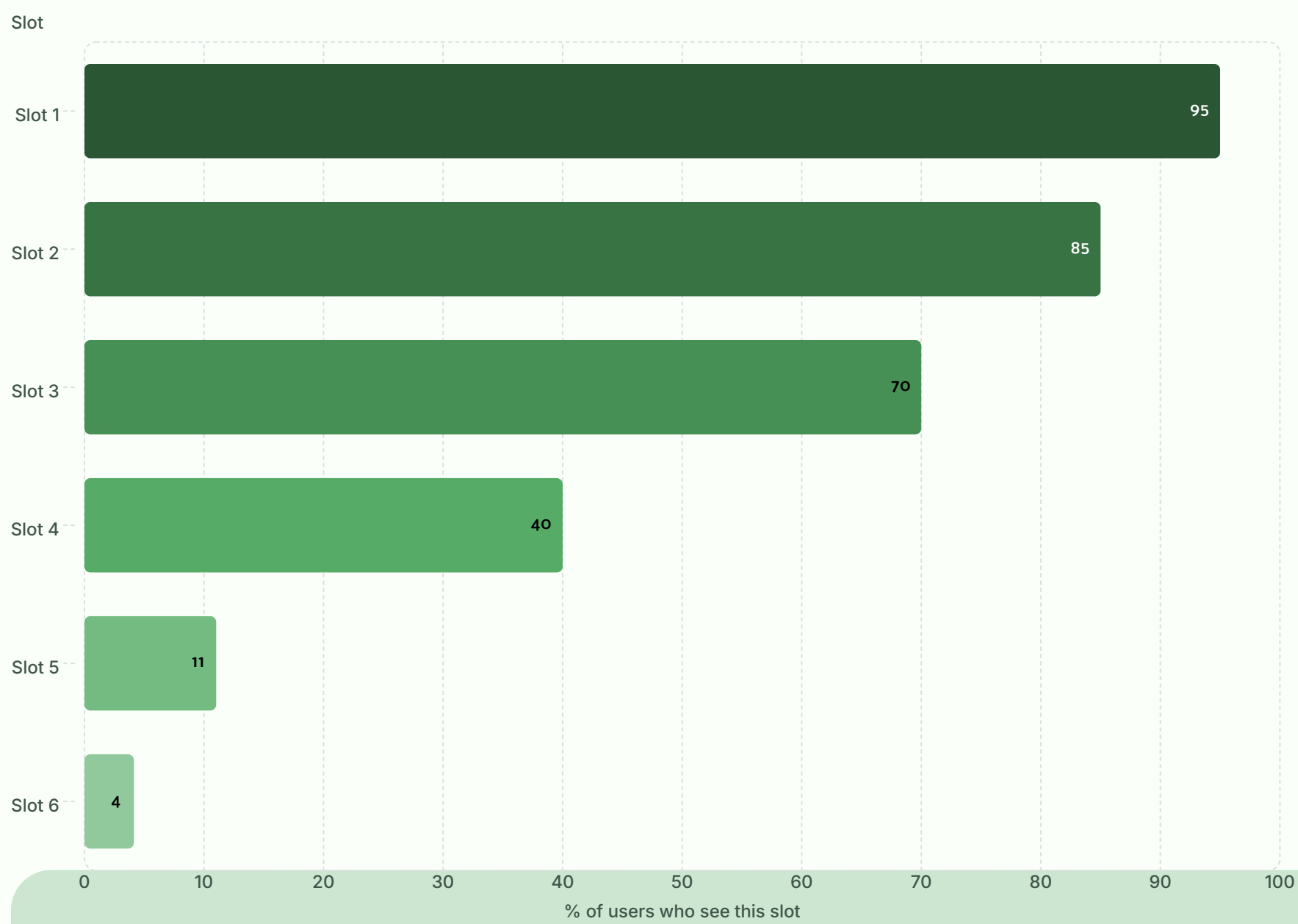
Product	Initial Score	Day-0 Rank
Item A	0.82	#1
Item B	0.74	#2
Item C	0.61	#3
Item D	0.55	#4
Item E	0.43	#5

⚠ Hidden assumption: **a click = the item is relevant**, regardless of position.

Made with GAMMA

Position decides who gets seen.

Users click higher results – not because those items are better, but because they appear first. The model cannot tell the difference.



The model only sees raw clicks. It **cannot separate item quality from slot advantage.**

Made with GAMMA

It learns the wrong thing — and scores keep improving.

The model accurately predicts future click rates. Every accuracy metric goes up. It is just very good at predicting the wrong thing.

What the model concludes

Item X at slot 1: **47 clicks**

Item Y at slot 8: **3 clicks**

Conclusion: **X ≫ Y**

What is actually true

Item X real quality: **0.62**

Item Y real quality: **0.61**

Conclusion: **X ≈ Y**

This is the hardest stage to catch. Nothing looks broken. Internal metrics are all green.

Made with GAMMA

Top items get stuck. New sellers starve.

With each retraining cycle the gap widens. Items that started high collect more clicks, score higher, rank higher – forever.

1 High Slot

2 More Clicks

3 Higher Score

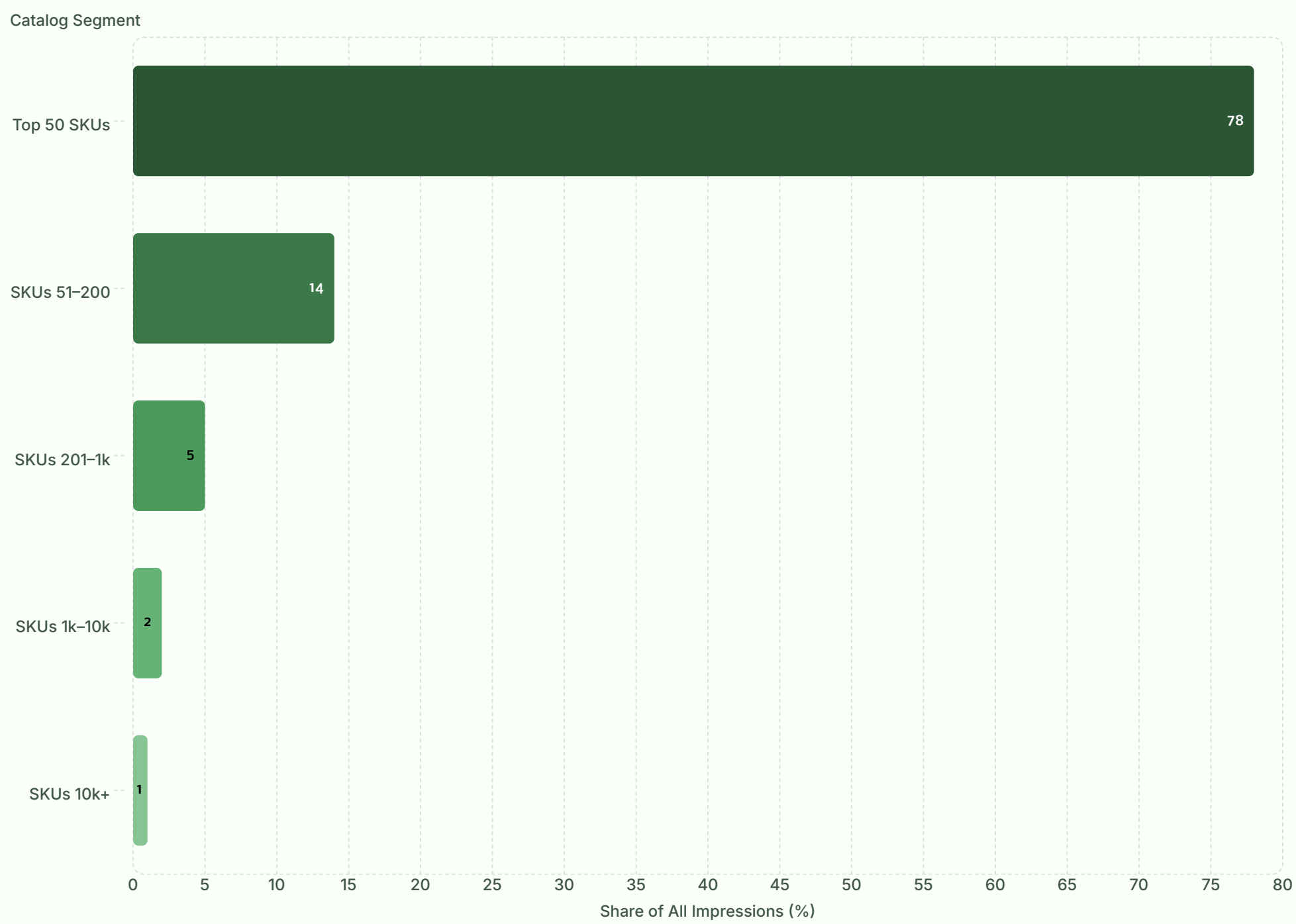
4 Higher Slot

A new seller lists 20 products. All land below slot 25. Zero orders. They churn in 30 days.
Starting slot = destiny. True quality stops mattering. Catalog diversity collapses.

Made with GAMMA

The catalog quietly collapses.

A platform with 10,000 listed products ends up showing the same 50 to almost everyone. Revenue may still grow – masking the problem for 12 - 18 months.



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How to catch it before revenue shows it?

Impression Concentration

Are a few SKUs grabbing all eyeballs?

⚠ **Rising Gini score → 60-90 day warning**

New Seller Zero-Impressions

Do new suppliers get any exposure in week 1?

⚠ **Exceeds 40% → 30-45 day warning**

Slot-Click Correlation

Is click rate basically just position rank?

⊗ **Correlation above 0.85 → 45-60 day warning**

Frustrated Exit Rate

Scrolled deep, clicked nothing, left.

⊗ **Rising above baseline → 15-30 day warning**

Made with GAMMA

Three interventions. All three, not one.

Solving any single one in isolation leaves the loop open.

1

Fix the Training Labels

3 - 6 MONTHS

Reweight clicks by how likely each slot was to be seen. Separates item quality from position advantage. **Treats the root cause.**

2

Reserve Impressions for New Items

4 - 8 WEEKS

Allocate 2 - 3% of slots to unproven products. **You cannot learn quality without some exposure.**

3

Cap Items per Seller per Page

1 - 2 WEEKS

Hard limit on how many items from one supplier appear together. **Buys time while the deeper fixes are built.**

Made with GAMMA

The loop can be broken.

The recommendation engine is not malicious – it is doing exactly what it was trained to do. The problem is what it was trained on.

Measure early

Track impression concentration, new seller exposure, and exit rates before revenue signals the damage.

Fix the signal

Position-corrected click labels break the feedback loop at its root – quality, not slot, drives rank.

Protect diversity

Impression reserves and seller caps keep the catalog alive while deeper fixes take hold.

A healthier ranking system means more sellers thrive, more products get discovered, and the platform grows sustainably – not just faster.

Made with GAMMA